AN AI-ENABLED INTELLIGENT STUDY ASSISTANT

INDIVIDUAL FINAL REPORT

Emotion and Progress-Aware Adaptive Study Reminder System

R26-SE-022

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DECLARATION

I declare that this individual final report presents my own contribution to the Smart Study Assistant project, specifically the Emotion and Progress-Aware Adaptive Study Reminder System. The implementation, analysis, testing evidence, interpretations, and conclusions reported in this document are presented for academic assessment and are the responsibility of the author.

Where academic literature, established algorithms, frameworks, or other external material has been used, the relevant sources are acknowledged through citations and the reference list. Any assistance from software tools, including generative AI used for language refinement or document organization, does not replace the author's responsibility for verifying the technical accuracy and originality of the submitted work.

Signature:  Date: 2026/10/11

My Signature: 

ACKNOWLEDGEMENT

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I also acknowledge the Faculty of Computing and the Department of Software Engineering of the Sri Lanka Institute of Information Technology for providing the academic environment and resources required to conduct this work.

I am grateful to my project teammates for their collaboration and for the integration of the individual components that together form the Smart Study Assistant. I also appreciate everyone who contributed feedback during the development and testing of the system.

ABSTRACT

This report presents the design and implementation of the Emotion and Progress-Aware Adaptive Study Reminder System, an individual component of the AI-Enabled Intelligent Study Assistant (R26-SE-022). The component addresses a limitation of conventional study reminder systems, which commonly schedule learning activities using fixed intervals or performance information without sufficiently considering the learner's current readiness.

The developed system combines learner readiness information, memory retention modelling, study progress, review priority, and learner interaction to support context-aware reminder decisions. A readiness service categorizes the learner into High, Medium, or Low readiness states and considers emotional states such as focused, neutral, confused, frustrated, and bored. Memory retention is estimated using an exponential forgetting formulation, while an SM-2 based review mechanism maintains review intervals, repetition counts, difficulty, and review history.

These inputs are combined by an adaptive reminder decision layer that can select an immediate reminder, a delayed reminder, or no reminder when the learner is not in a suitable state. The system also recommends an appropriate study activity, such as active recall, guided review, reading notes, or a short break. A backend reminder service, MongoDB storage, authenticated APIs, a frontend reminder queue, browser notification support, configurable reminder preferences, and learner feedback handling were integrated into the implementation.

Functional testing was conducted using predefined scenarios covering readiness states, review priority, unfamiliar topics, fresh topics, review-interval updates, and scheduling decision boundaries. The results demonstrate the functional integration of the main reminder workflow. The implementation is presented as a functionally tested prototype; quantitative claims about long-term learning improvement or user-study effectiveness are not made without corresponding empirical evidence.

Keywords: Adaptive Study Reminders; Emotion-Aware Learning; Spaced Repetition; Memory Retention; Learner Readiness; Personalized Learning; Smart Study Assistant.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
DSR	Design Science Research
HTTP	Hypertext Transfer Protocol
ML	Machine Learning
NASA-TLX	NASA Task Load Index
REST	Representational State Transfer
SLIIT	Sri Lanka Institute of Information Technology
SM-2	SuperMemo Algorithm 2
SUS	System Usability Scale
UI	User Interface
UX	User Experience

1. INTRODUCTION

1.1 Background and Context

University students are required to manage increasing amounts of academic content, deadlines, assessments, and independent study. Digital learning tools can assist learners by organizing study material and reminding them when learning activities are due. However, a reminder that is appropriate from a purely time-based perspective may not always be appropriate from the learner's perspective.

Spaced repetition and memory-based review systems provide a useful foundation for deciding when study material should be revisited. Nevertheless, these approaches primarily focus on learning performance and review history. They do not necessarily account for the learner's current emotional or cognitive readiness when a reminder is delivered.

The broader Smart Study Assistant project addresses this problem by combining multiple intelligent learning-support components. The individual component presented in this report focuses specifically on the scheduling and notification layer. Its purpose is to use learner readiness and learning progress to determine whether a study reminder should be presented and what type of study activity should be recommended.

The project proposal identifies the scope of this component as determining when, how often, and in what content modality a study reminder should be delivered given the learner's current state. The component is therefore positioned between learner-state information, learning-progress information, and the reminder-delivery interface.

1.2 Problem Statement

Conventional reminder systems commonly use fixed schedules or performance-based review intervals. Such systems can recommend that a learner study at a time when the learner is confused, frustrated, bored, or otherwise not ready to engage. Conversely, a learner may be highly prepared to study while an important topic is approaching a low-retention state, yet a fixed schedule may not take advantage of that opportunity.

The problem addressed by this research is therefore the lack of a study reminder mechanism that jointly considers learner readiness and learning progress when determining reminder timing and activity type. The component aims to reduce unnecessary study interruptions while maintaining attention on material that has a higher review priority.

1.3 Research Scope

The research scope is limited to the adaptive study reminder component of the Smart Study Assistant. The component receives learner-state information from the wider platform, processes study-item and review information, determines reminder priority, selects an appropriate reminder action, recommends a study activity, and delivers or exposes the reminder through the web application.

The component does not replace the sibling emotion-detection module. Instead, it consumes the readiness-related information required for scheduling. Similarly, the component does not attempt to implement the entire Smart Study Assistant; it provides the reminder and review-management functionality assigned to the individual contribution.

1.4 Literature Review

Spaced repetition is a widely used approach for improving long-term retention by scheduling repeated exposure to learning material at increasing intervals. The SM-2 algorithm provides a practical basis for adjusting review intervals using previous review performance [1]. In this project, an SM-2 based review mechanism is used to maintain study-item review state and identify material that requires additional attention.

Affective computing provides the broader theoretical basis for treating emotional information as a meaningful input to interactive systems. Picard, Vyzas, and Healey established important foundations for using affective signals in computing systems [2]. Within educational applications, this concept supports the idea that learner state can be considered alongside conventional learning-performance data.

Cognitive Load Theory emphasizes the limitations of working memory and the effects of excessive cognitive demand on learning [3]. This provides a theoretical motivation for avoiding unnecessarily demanding reminders when a learner is experiencing low readiness. The reminder system therefore considers readiness when deciding whether to issue an immediate study prompt.

Recent intelligent tutoring and personalized-learning research demonstrates the movement from static educational software toward systems that adapt content and interactions to learner characteristics [4]–[7]. These studies provide the broader research context for personalization, while the present work concentrates specifically on adaptive reminder timing and study activity selection.

Personalized reminder systems have also been studied in mobile learning contexts, demonstrating the importance of contextual reminder delivery rather than treating notifications as isolated events [8]. The present system extends this direction by integrating learner readiness with memory-based review priority.

1.5 Research Gap

The project proposal identifies a gap between performance-driven scheduling and affective learner-state information. Existing spaced-repetition approaches primarily use performance and review history, while affective-computing approaches can identify learner state without necessarily connecting that state to reminder scheduling. The proposed component addresses this gap by connecting readiness information with memory-based review priority and reminder delivery.

Dimension	Existing approach	Gap addressed by this component
Scheduling basis	Fixed intervals or performance history	Considers learner readiness together with review priority
Affective integration	Emotion detection often operates separately	Uses readiness information as a scheduling input
Reminder content	Often static or predefined	Recommends activity type according to learner state
Adaptiveness	Fixed or rule-based scheduling	Uses context-sensitive decision rules for send, delay, or skip
Feedback	Reminder response may be treated independently	Records accept, snooze, and dismiss interactions

1.6 Research Value

The research value of the component is the design and implementation of a reminder decision workflow that connects learner readiness with memory-based review priority. Rather than assuming that the mathematically ideal review time is always the ideal notification time, the system introduces learner-state information into the decision process.

1.7 Product Value

The product value lies in providing a study assistant that can reduce the rigidity of conventional reminder systems. The component can support students by reducing unnecessary prompts during low-readiness states, prioritizing material that requires review, and recommending a suitable study activity. The proposal also identifies higher-education institutions and learning platforms as potential deployment contexts.

2. OBJECTIVES

2.1 Main Objective

To design and implement an Emotion and Progress-Aware Adaptive Study Reminder System that integrates learner readiness with learning progress and memory-retention information to generate personalized study reminders while reducing unnecessary cognitive demand.

2.2 Specific Objectives

1. Model study-item memory retention and review priority using an SM-2 based review mechanism and forgetting formulation.
2. Integrate learner-state information supplied by the wider Smart Study Assistant into a readiness assessment with High, Medium, and Low readiness categories.
3. Develop adaptive reminder decision rules that select an immediate reminder, delayed reminder, or skipped reminder according to readiness and review priority.
4. Recommend a study activity appropriate to learner readiness and study-item characteristics.
5. Implement a backend reminder service with authenticated retrieval, storage, feedback handling, and review-state updates.
6. Implement a web-based reminder queue and notification workflow with configurable reminder preferences.
7. Validate the functional behavior of the reminder system through predefined test scenarios and boundary-condition testing.

2.3 Objective-to-Implementation Mapping

Objective	Implemented evidence
Memory retention and priority	Retention probability, forgetting risk, review priority, SM-2 state
Readiness assessment	High / Medium / Low readiness and supported learner states
Adaptive reminder decisions	SEND_NOW / DELAY / SKIP decision layer
Activity recommendation	Active recall, guided review, reading notes, short break
Backend service	FastAPI endpoints, authenticated reminder retrieval, MongoDB
Frontend delivery	Reminder queue, settings, toast/browser notification support
Feedback	Accepted, Snoozed, Dismissed responses and review updates
Testing	Unit/functional tests for reminder scenarios and decision boundaries

3. METHODOLOGY

3.1 Research Approach

The project follows a Design Science Research approach. DSR is appropriate because the research is centered on designing, implementing, and evaluating a software artefact. The process consists of identifying the practical problem, reviewing relevant knowledge, designing the solution, implementing the artefact, and evaluating its behavior.

For this component, the DSR process was applied iteratively. Requirements and literature were used to design the reminder workflow, the workflow was implemented as modular services, functional tests were created around expected learner scenarios, and the results were used to identify limitations and future improvements.

3.2 System-Level Method

The reminder component operates as part of the broader Smart Study Assistant. The component receives learner-state and learning-progress information, transforms these inputs into scheduling features, evaluates review priority, selects a reminder action, recommends an activity, and exposes the resulting reminder through the application.

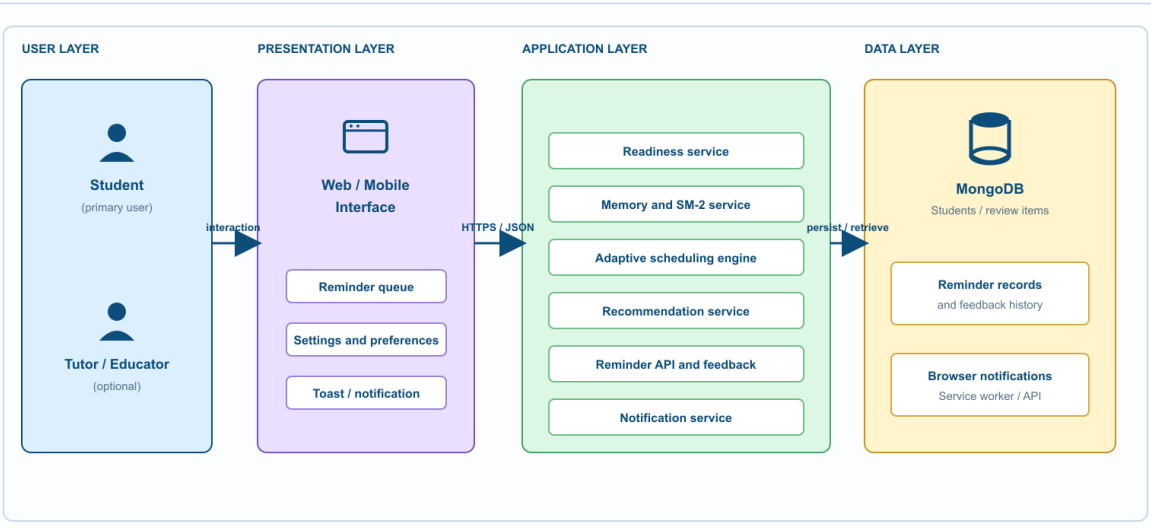


Figure 1: OVERALL SMART STUDY ASSISTANT ARCHITECTURE

3.3 Technical Methods and Algorithms

The main technical methods used in the implemented component are readiness assessment, exponential memory-retention estimation, SM-2 based review management, rule-based adaptive scheduling, activity recommendation, feedback processing, and reminder delivery.

3.4 Learner Readiness Assessment

The readiness service produces High, Medium, or Low readiness. The implementation also represents learner states including focused, neutral, confused, frustrated, and bored. Where the trained readiness artifact is available, the service can use its prediction. A deterministic fallback is available to keep the pipeline operational when the artifact is unavailable.

The readiness result is treated as a scheduling input rather than as an academic judgement. The intention is to estimate whether a learner is currently in a suitable state to receive a particular study demand.

3.5 Memory Retention Modelling

The implemented retention calculation uses an exponential forgetting formulation:

$$R(t) = e^{(-t/S)}$$

Where R(t) is the estimated retention probability, t is the elapsed time since the previous review, and S represents memory stability.

A review-priority value is derived from retention as:

$$\text{Priority} = (1 - R(t)) \times 100$$

A higher priority indicates greater forgetting risk and therefore a stronger reason to consider a review reminder.

3.6 SM-2 Based Review Management

The review-management service maintains repetition count, review interval, easiness factor, difficulty, review history, and next review date. Successful reviews can increase the interval between reviews, while lower review grades can reduce or reset the review sequence. This allows the reminder system to represent the learner's progress at study-item level.

3.7 Adaptive Reminder Decision

The reminder decision layer combines readiness and review priority. The principal actions are SEND_NOW, DELAY, and SKIP. High readiness combined with significant review need can result in an immediate reminder. Moderate readiness can result in a delayed reminder, while low readiness can suppress or postpone non-critical study demands.

Condition	Typical action	Rationale
High readiness + high review priority	SEND_NOW	Learner is prepared and the material requires attention
Medium readiness + sufficient priority	SEND_NOW or DELAY	Review is relevant but timing can be moderated
Medium readiness + lower urgency	DELAY	Avoid unnecessary immediate interruption
Frustrated or bored state	DELAY	Reduce immediate cognitive demand
Low readiness + non-critical item	SKIP / DELAY	Avoid unnecessary study pressure

3.8 Decision Explanation

Each scheduling decision can include an explanation containing readiness interpretation, forgetting risk, priority, and the reason for the selected action. This makes the automated reminder decision more understandable to developers, evaluators, and learners.

3.9 Study Activity Recommendation

Learner/context condition	Recommended activity
High readiness and focused state	Active recall
Confused or unfamiliar topic	Guided review
Difficult topic with medium readiness	Reading notes
Frustration or boredom	Short break
Low readiness	Reduced study demand / delay

3.10 Reminder Data and Backend Processing

The backend stores reminder information such as topic, content type, readiness tier, readiness score, retention probability, activity type, scheduled time, delivery time, response time, and decision reason. Authenticated access is used so that reminder retrieval is associated with the correct learner.

3.11 Feedback Processing

Learner feedback is represented using Accepted, Snoozed, and Dismissed states. The backend records the reminder response and can update the related review item using the learner's submitted review grade. This creates a feedback path between reminder interaction and study-item review state.

3.12 Data and Ethical Considerations

The proposal defines anonymized interaction logs, readiness-related information, and self-reported mood information as potential research data sources. It also specifies consent, withdrawal rights, secure storage, and avoidance of raw biometric-data storage as privacy principles.

For the implemented reminder component, privacy should be maintained by minimizing the information persisted by the scheduling layer, applying authenticated access, and avoiding unnecessary storage of raw emotion-related source data. Ethical claims about a completed participant study should only be made if the corresponding study was actually conducted.

3.13 Validation Method

The current implementation was validated primarily through functional and scenario-based testing. Tests were designed to verify whether the reminder decision logic, review-state updates, activity recommendations, and reminder interactions behave as specified. The proposal's planned user-study metrics, such as adherence and SUS, are treated as evaluation targets rather than completed results unless supported by actual collected data.

4. PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The design requirements were derived from the problem definition, project proposal, wider Smart Study Assistant architecture, and implemented functionality.

4.1.1 Functional Requirements

ID	Requirement	Description
FR1	Memory retention and review priority	Calculate retention and identify study items requiring review.
FR2	Readiness assessment	Determine High, Medium, or Low learner readiness.
FR3	Adaptive reminder decision	Select SEND_NOW, DELAY, or SKIP based on context.
FR4	Activity recommendation	Recommend an appropriate study activity.
FR5	Reminder delivery	Expose reminders through the web queue and notification interface.
FR6	Feedback collection	Record Accepted, Snoozed, and Dismissed responses.
FR7	Review-state update	Update related review information following learner feedback.
FR8	Preference management	Support reminder frequency and quiet-hour preferences.
FR9	Authenticated access	Restrict reminder records to the authenticated learner.

4.1.2 Non-Functional Requirements

Category	Requirement
Performance	Reminder decisions should be generated efficiently under normal application load.
Reliability	The system should continue operating using defined fallback behavior when optional model artifacts are unavailable.
Usability	Reminder interaction should require minimal user effort.
Privacy	Learner-state and interaction information should be minimized and protected.
Security	Authenticated API access should prevent unauthorized reminder retrieval or modification.
Maintainability	Scheduling, memory, reminder, and interface functionality should remain modular.
Scalability	The design should allow the reminder service to be deployed independently as the wider platform grows.

4.2 Component Architecture

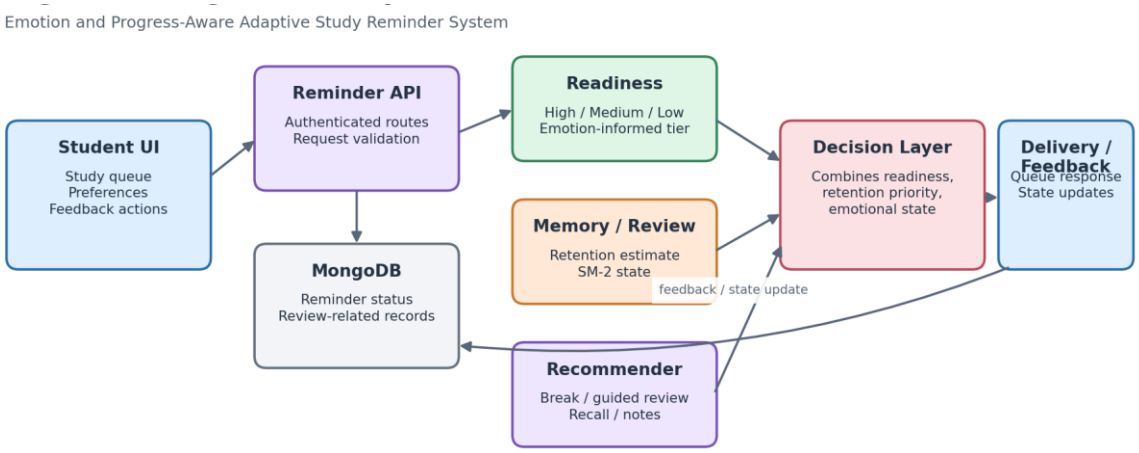


Figure 2: INDIVIDUAL COMPONENT ARCHITECTURE

The component architecture separates readiness assessment, memory/review management, scheduling decisions, activity recommendation, reminder persistence, and user interaction. This separation allows each responsibility to be tested independently and reduces coupling between the scheduling logic and the presentation layer.

4.3 Component Interactions

1. The system retrieves or creates the study-item information required for the reminder decision.
2. The readiness service determines the learner's current readiness state.
3. The memory service estimates retention and review priority.
4. The scheduler evaluates readiness and priority to determine SEND_NOW, DELAY, or SKIP.
5. The recommendation service selects an appropriate study activity.
6. The reminder record is created or retrieved through the reminder service.
7. The frontend displays the reminder through the queue or notification interface.
8. The learner responds to the reminder.
9. The backend records the feedback and updates the related review state where applicable.

4.6 Data Flow

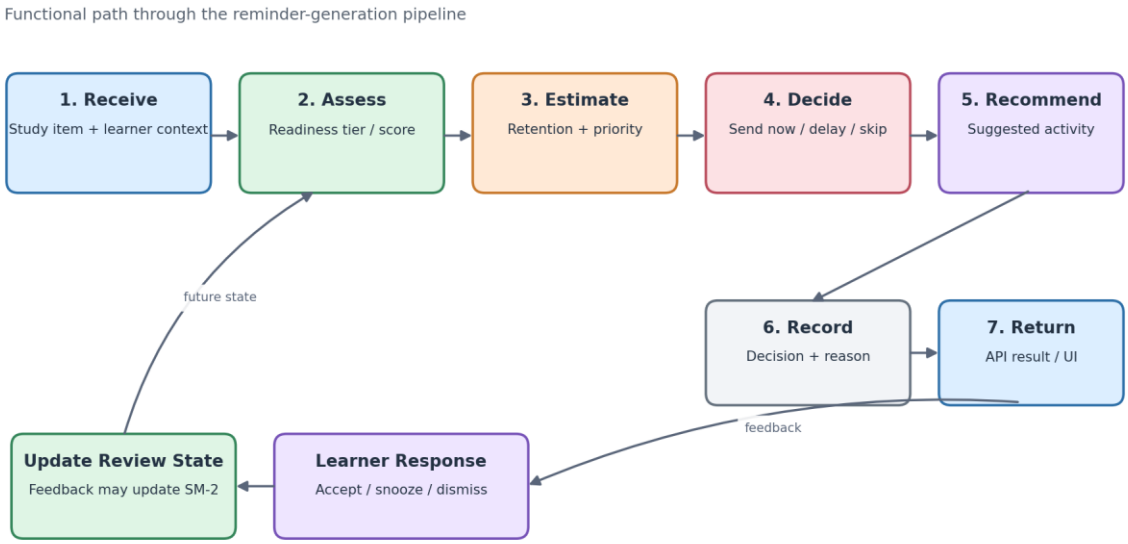


Figure 3: DATA FLOW DIAGRAM

4.5 API Design

Endpoint	Method	Purpose
/api/reminder/generate	POST	Run the unified reminder decision workflow.
/api/v1/reminders	GET	Retrieve reminders for the authenticated learner.
/api/v1/reminders/{reminder_id} /feedback	POST	Submit reminder feedback.
/api/reminder/preferences	GET/POST	Retrieve or update reminder preferences.
/api/schedule	POST / related route	Manual scheduling invocation / scheduling workflow entry point.

4.6 Reminder Data Model

Field	Purpose
Reminder ID	Unique reminder identifier
Topic title	Study item associated with the reminder
Content type	Type of recommended study activity
Readiness tier	High, Medium, or Low
Readiness score	Readiness information used by the scheduler
Retention probability	Estimated memory retention
Activity type	Recommended learner action
Scheduled time	Intended reminder time
Delivery time	Actual delivery/exposure time when available
Response time	Learner interaction time
Decision reason	Explanation for the scheduling decision
Status	Current reminder state

4.7 User Workflow

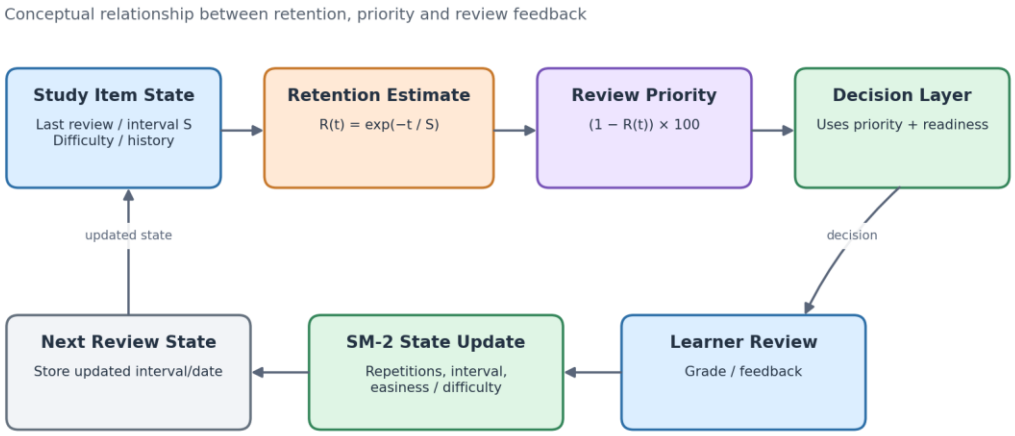


Figure 4: USER WORKFLOW

4.8 Design Decisions

A modular service design was selected so that the reminder component can interact with the wider Smart Study Assistant without tightly coupling every responsibility. A rule-based scheduling layer was retained because it is transparent, testable, and suitable for the current implemented prototype.

The design also separates reminder timing from activity recommendation. This means the system can first determine whether a reminder is appropriate and then determine what the learner should do, rather than treating reminder timing and study content as the same decision.

4.9 Security and Privacy Design

The implementation uses authenticated reminder retrieval and learner-specific filtering. The reminder service returns records associated with the authenticated learner, reducing the risk of cross-user access. The system also follows a data-minimization principle by avoiding unnecessary storage of raw emotion-related source data within the reminder layer.

The proposal identifies HTTPS/TLS, encrypted storage, role-based access, anonymization, and privacy-preserving handling of readiness information as system-level security requirements. These should be described as implemented only where verified in the final deployed environment.

5. IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

Category	Technology	Purpose
Backend	Python / FastAPI	Reminder APIs and processing services
Frontend	React / Next.js	Reminder queue and interaction interface
Database	MongoDB	Reminder and interaction storage
Memory model	SM-2 based implementation	Review scheduling and progress state
ML / readiness	scikit-learn / readiness service	Readiness prediction and fallback logic
Notifications	Web Notification API / Service Worker	Browser notification support
Testing	Python test suite	Functional and decision-rule verification
Version control	Git / GitHub	Source management and collaboration

5.2 Implemented Modules

Module	Main responsibility
readiness_service.py	Learner readiness prediction and readiness categorization
memory_service.py	Retention and memory-related processing
sm2.py	Review interval and study-item review state
adaptive_scheduler.py	Adaptive reminder decision rules
scheduler_service.py	Additional scheduling-rule orchestration
recommendation_service.py	Study activity recommendation
unified_pipeline.py	Integrated reminder-generation workflow
reminders.py / reminder API	Reminder retrieval and feedback handling
reminder_preferences.py	Reminder frequency and quiet-hour preferences
ReminderToast.js	Frontend reminder display and interaction
sw.js	Browser service-worker notification support

5.3 Unified Reminder Workflow

The unified reminder-generation workflow retrieves or creates study-item information, predicts readiness, calculates retention and review priority, determines the reminder action, recommends a study activity, creates an event record, and returns the decision with supporting reasoning. This provides a single workflow through which the major individual functions can be exercised.

5.4 Frontend Implementation

The frontend provides due-today, upcoming, and available reminder views, refresh functionality, reminder settings, frequency configuration, quiet-hour configuration, notification permission handling, push-subscription support, and feedback submission.

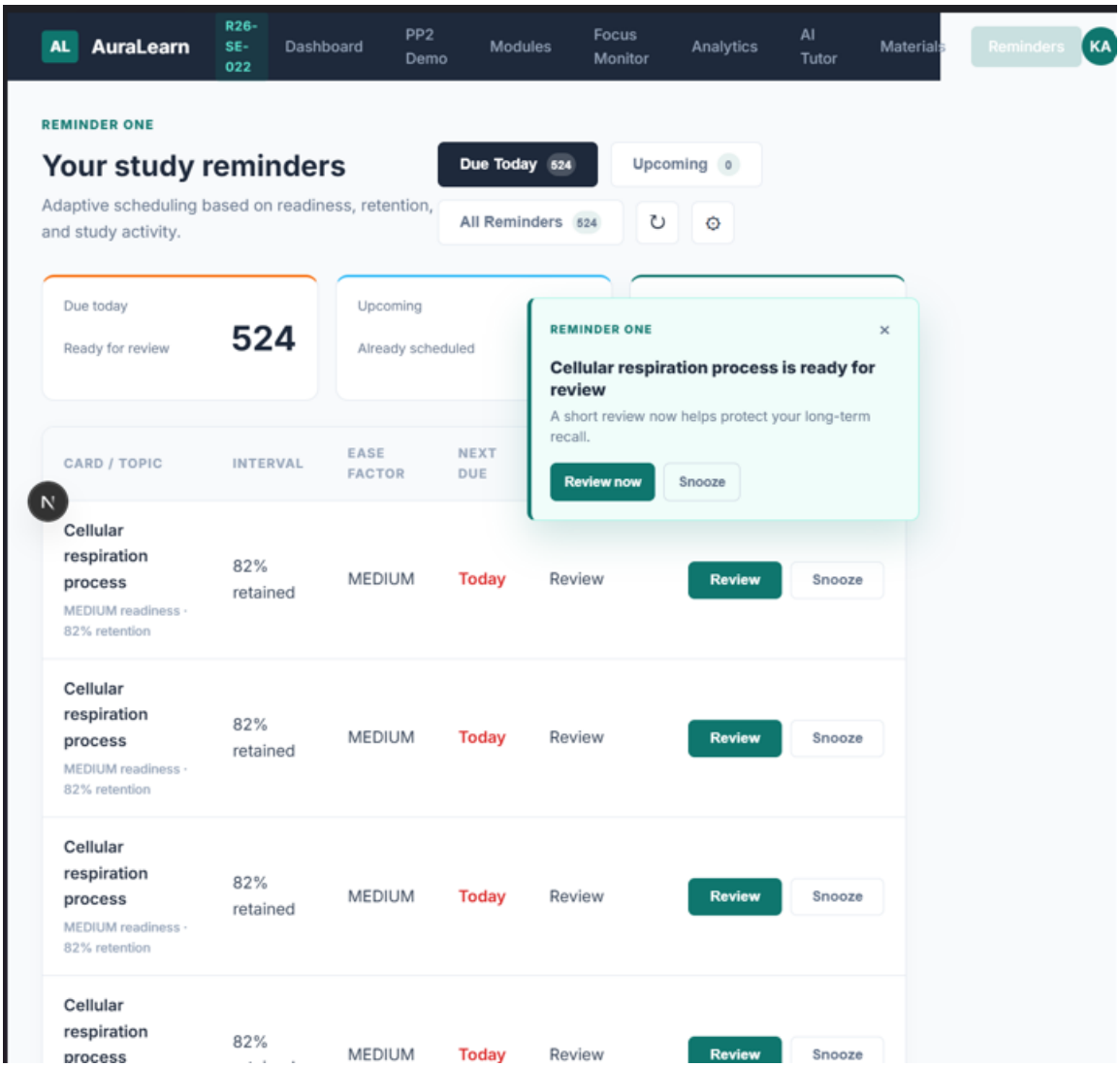


Figure 5: REMINDER QUEUE SCREENSHOT

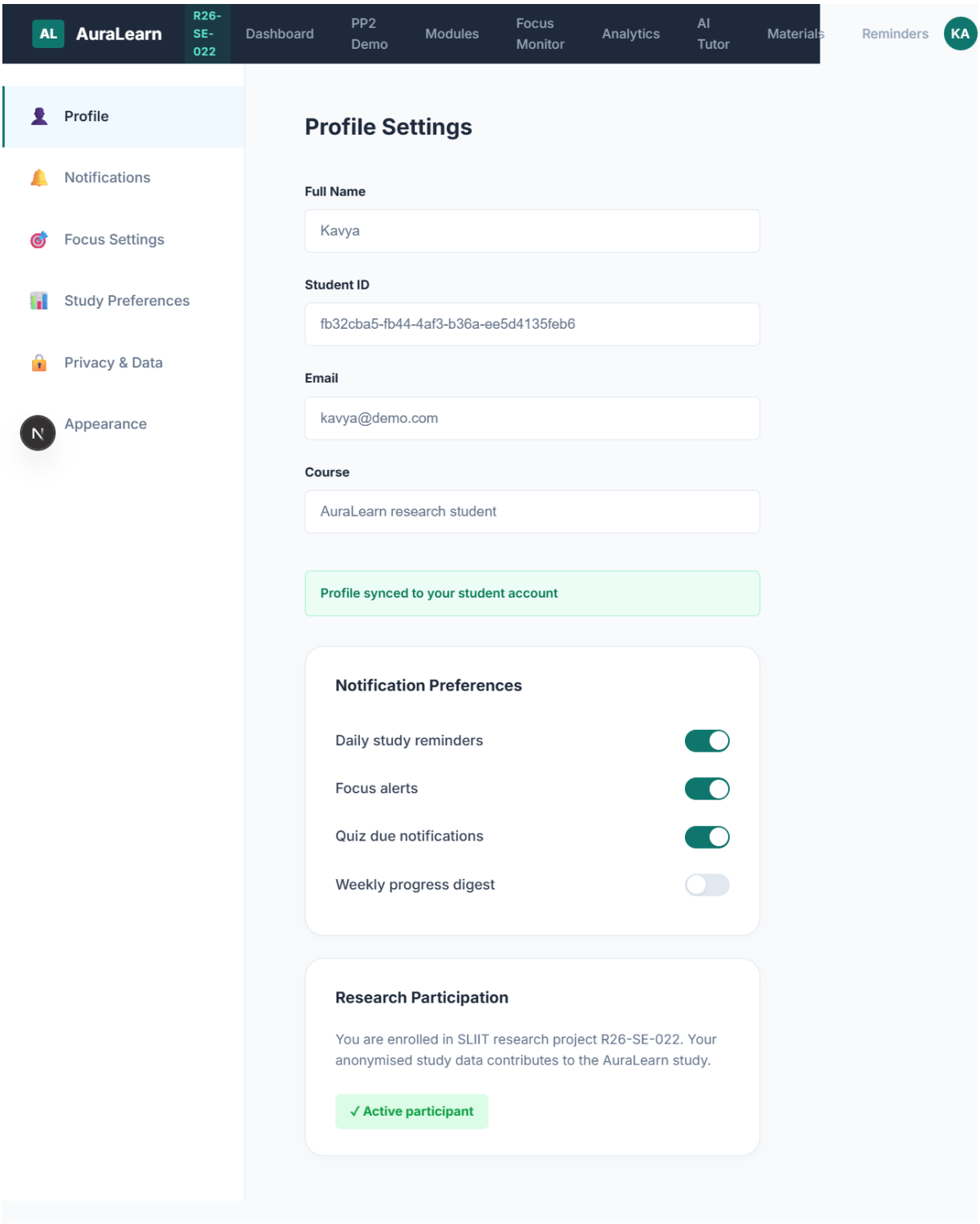


Figure 6: REMINDER SETTINGS SCREENSHOT

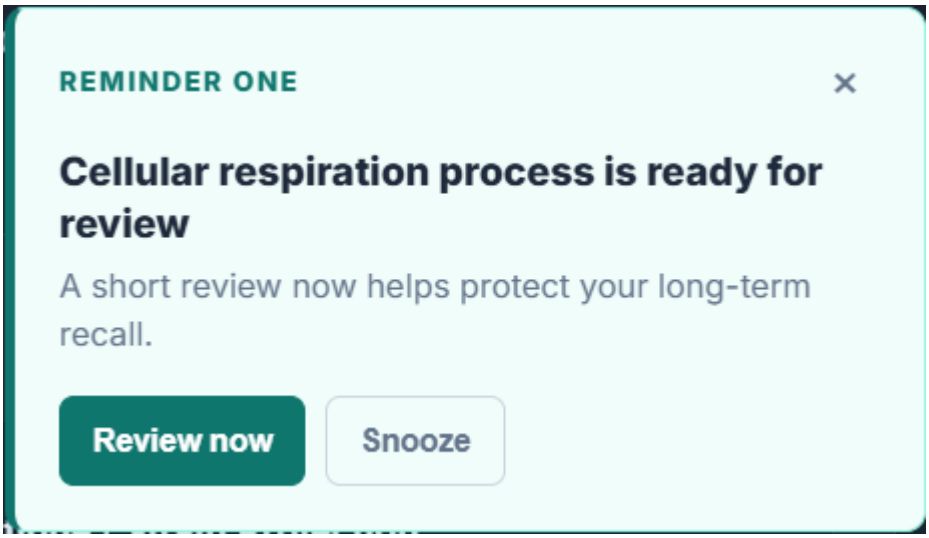


Figure 7: REMINDER TOAST / NOTIFICATION SCREENSHOT

5.5 Feedback Workflow

When a learner accepts, snoozes, or dismisses a reminder, the frontend submits the response to the backend. The backend identifies the reminder for the authenticated learner, records the response and response time, and updates the related review item when the necessary review information is available.

5.6 Functional Testing

Test case	Expected behavior	Status
High-readiness review	Important review receives an immediate study decision	PASS – verified functional scenario
Frustration / low readiness	Immediate study demand is reduced through delay/skip behavior	PASS – verified functional scenario
Unfamiliar topic	Guided review is recommended	PASS – verified functional scenario
Fresh topic	Low review urgency does not create unnecessary priority	PASS – verified functional scenario
SM-2 interval update	Review interval changes according to review outcome	PASS – verified functional scenario
Default thresholds	Scheduler uses configured default decision boundaries	PASS – verified
Custom thresholds	Scheduler responds to configured threshold changes	PASS – verified
Decision boundaries	Send/delay/skip boundaries behave consistently	PASS – verified
Readiness label case handling	Equivalent readiness labels are handled consistently	PASS – verified
Low-readiness protection	Low readiness does not create unnecessary immediate demand	PASS – verified

5.7 Testing Evidence

The project test suite includes targeted tests for reminder decisions and the PP2 audit scenarios. The tests demonstrate functional correctness of predefined cases; they should not be interpreted as evidence of statistical model accuracy or learning improvement.

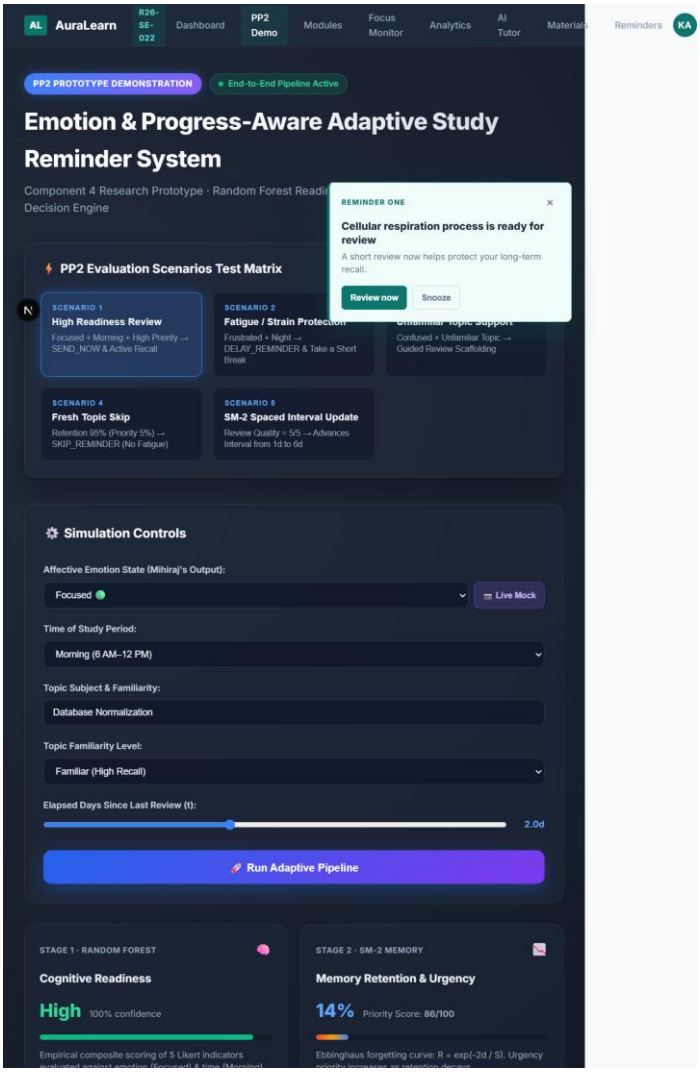


Figure 8: TEST EXECUTION SCREENSHOT

5.8 Integration Testing

Integration testing focuses on the movement of information between readiness assessment, memory/review services, scheduling logic, recommendation, reminder storage, and frontend interaction. The integrated pipeline provides a single decision output containing the action, supporting information, and explanation.

5.9 Implementation Limitations

The current implementation is a functional prototype. The complete automatic background scheduling tick is not fully operational, and some reminder records can be seeded or generated for demonstration. Browser push delivery also depends on appropriate notification configuration. These limitations mean that the system should not be presented as a fully autonomous production deployment.

6. RESULTS AND VALIDATION

6.1 Functional Results

The implemented workflow successfully connects readiness assessment, retention calculation, reminder prioritization, activity recommendation, reminder storage, frontend presentation, and learner feedback. The functional scenarios verify that the main decision rules respond appropriately to different learner conditions and review priorities.

The strongest evidence currently available is software-level functional verification. The project should not report numerical improvement in adherence, retention, cognitive workload, or usability unless corresponding user-study data and analysis are available.

6.2 Readiness and Scheduling Behaviour

Scenario	Readiness / context	Expected system response
Focused learner with important review	High readiness + high priority	Send now
Moderate readiness with moderate urgency	Medium readiness	Send or delay according to priority
Frustrated learner	Low/unfavourable emotional state	Delay or reduce demand
Bored learner	Low/unfavourable state	Delay or reduce demand
Unfamiliar difficult topic	Confusion + unfamiliarity	Guided review
Well-established familiar topic	High readiness + familiarity	Active recall

6.3 Memory and Review Results

The memory component maintains the review state required to calculate retention and update review intervals. The retention formulation provides a continuously decreasing estimate as time passes after a review, while SM-2 state variables provide a mechanism for adjusting the interval according to review outcomes.

This combination allows the reminder system to distinguish between material that is newly introduced, material that has been successfully reviewed, and material that has become increasingly important to revisit.

6.4 Reminder Interface Results

The implemented frontend provides a learner-facing reminder queue and notification workflow. Learners can view reminder information, access settings, and respond through supported actions. Reminder status is updated after interaction, allowing the visible queue to reflect the learner's latest response.

6.5 Testing Results

The predefined functional scenarios and scheduling-boundary tests demonstrate that the implemented decision logic behaves according to the specified rules. The tests cover both typical cases and boundary conditions, increasing confidence that changes in readiness and priority produce predictable reminder decisions.

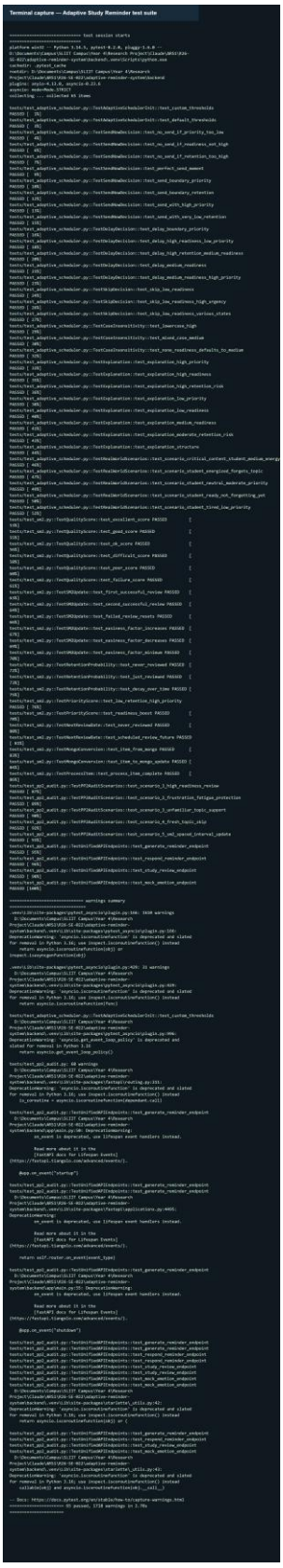


Figure 9: REMINDER QUEUE RESULT

6.6 Validation Against Proposed Research Metrics

The original proposal defines adherence rate, retention improvement, NASA-TLX cognitive fatigue, false-positive reminder rate, SUS, and readiness-score alignment as evaluation metrics. These are useful for a future controlled user evaluation. However, the current implementation evidence available for this report is primarily functional, so these metrics are not presented as achieved experimental results.

Metric	Proposal target	Current report status
Adherence rate	>85% improvement vs. baseline	Not claimed; requires controlled user data
Retention improvement	Statistically significant improvement	Not claimed; requires pre/post evaluation
NASA-TLX	Lower cognitive fatigue than baseline	Not claimed; requires participant responses
False-positive reminder rate	<15%	Not claimed without measured reminder outcomes
SUS	>70	Not claimed without completed questionnaire
Readiness alignment	>80%	Not claimed without validated labels/self-report comparison

6.7 Validation Conclusion

The current validation establishes that the implemented reminder component can execute its primary decision and interaction workflows under predefined scenarios. Further empirical evaluation is required to establish whether the system improves real learner adherence, retention, wellbeing, or usability.

7. DISCUSSION

7.1 Interpretation of Findings

The main finding from the implementation is that learner readiness and memory-based review priority can be represented within a single reminder workflow. A conventional review algorithm can identify when material is likely to require attention, while the readiness component provides additional context about whether the learner is currently suitable for that study demand.

The adaptive decision layer provides a practical bridge between these two dimensions. It prevents the reminder mechanism from treating review priority as the only factor determining delivery. For example, high review priority can justify an immediate reminder when readiness is high, whereas the same priority can be delayed when the learner is experiencing a low-readiness state.

7.2 Relationship to Existing Approaches

The system retains the strengths of spaced repetition by maintaining item-level review state and memory-based priority. At the same time, it extends the reminder workflow with learner-state information and activity recommendation. This provides a more context-sensitive design than a reminder mechanism that only considers elapsed time or previous quiz performance.

The system is therefore best interpreted as a hybrid intelligent reminder workflow rather than as a replacement for established spaced-repetition methods.

7.3 Research Contribution

The research contribution is the design and implementation of an integrated reminder workflow that connects learner readiness with memory retention and study progress. The component operationalizes this concept through readiness categories, retention probability, review priority, decision rules, activity recommendation, and feedback handling.

7.4 Product Contribution

The product contribution is an adaptive reminder feature that can be integrated into a wider intelligent study assistant. It adds contextual timing, personalized activity recommendation, reminder preferences, and feedback interaction to the study-planning experience.

7.5 Limitations and Threats to Validity

- The current evidence is primarily functional rather than a completed controlled user study.
- The readiness component depends on the quality of upstream learner-state information.
- Retention estimates are model-based approximations and may not perfectly represent every learner or subject.
- Some reminder records can be seeded or generated for demonstration.
- The complete continuous background scheduling process requires further integration.
- Browser push delivery depends on browser permission and deployment configuration.
- Functional test success does not establish improved academic outcomes.

7.6 Implications

The implementation suggests that adaptive study reminders can be designed around both learning urgency and learner readiness. This can be useful in environments where the objective is not simply to maximize the number of reminders, but to make reminders more contextually appropriate and less intrusive.

8. COMMERCIALISATION AND SUSTAINABILITY

8.1 Target Users and Market

The primary target users are university students and learners using digital study platforms. Institutional users can include universities, private education providers, and organizations delivering structured training. The component can be offered as a feature within a broader intelligent learning platform or as an independently deployable reminder service.

The original proposal identifies higher-education institutions, learning platforms, and student-support environments as potential deployment contexts.

8.2 Value Proposition

- Context-aware study reminders rather than purely fixed-time notifications.
- Integration of memory-based review priority with learner readiness.
- Personalized study activity recommendations.
- Configurable reminder frequency and quiet hours.
- Learner feedback through accept, snooze, and dismiss interactions.
- Modular API-based design suitable for integration with larger learning platforms.

8.3 Potential Business Models

Model	Description
Institutional SaaS	Universities or education providers subscribe based on student usage.
Freemium	Basic reminders are free, with advanced personalization and analytics offered as premium features.
Institutional licensing	Organizations license the system for controlled internal deployment.
API service	Third-party learning platforms integrate reminder functionality through APIs.

8.4 Sustainability

The component can be sustained through modular deployment, reuse of open-source technologies, and integration with existing educational platforms. Long-term sustainability would require monitoring system performance, maintaining the data pipeline, validating reminder effectiveness, and updating the implementation as learner needs and browser notification technologies evolve.

8.5 Intellectual Property Considerations

Potential intellectual-property considerations include the software implementation, integration design, and project-specific decision workflow. Any formal intellectual-property claim should be made according to SLIIT policies and should not be stated as registered unless registration has actually occurred.

8.6 Sustainable Development Consideration

The project can be aligned conceptually with Sustainable Development Goal 4 (Quality Education) because it aims to support effective and personalized learning. This is an application-level alignment rather than a measured SDG impact claim. Any quantified impact should be supported by future educational-outcome evidence.

9. CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

The Emotion and Progress-Aware Adaptive Study Reminder System was developed as an individual component of the Smart Study Assistant to provide personalized study reminders based on learner readiness and learning progress. The component combines readiness assessment, memory retention modelling, SM-2 based review management, review prioritization, adaptive reminder decisions, activity recommendation, reminder delivery, preferences, and feedback.

The implemented workflow demonstrates that reminder timing can be treated as a contextual decision rather than as a fixed notification event. When a study item requires attention and the learner is ready, the system can provide an immediate reminder. When the learner is not in a suitable state, the system can delay or suppress non-critical study demands.

The component was integrated with a backend API, MongoDB reminder storage, frontend queue, browser notification support, and feedback processing. Functional testing verified important reminder scenarios and decision boundaries. The resulting prototype provides a foundation for further empirical research and production-level development.

9.2 Achievement of Objectives

Objective	Achievement
Memory retention and review priority	Implemented
Learner readiness assessment	Implemented
Context-sensitive reminder decisions	Implemented and functionally tested
Study activity recommendation	Implemented
Reminder backend and storage	Implemented
Frontend reminder workflow	Implemented
Preferences and notification support	Implemented
Feedback and review-state update	Implemented
Empirical user-study validation	Future evaluation / not claimed

9.3 Individual Research Contribution

The individual contribution covers the design and implementation of the Emotion and Progress-Aware Adaptive Study Reminder System. The contribution includes readiness-aware reminder decision logic, memory and review prioritization, study activity recommendation, reminder API and storage integration, frontend reminder interaction, preferences, feedback processing, and functional testing.

9.4 Future Directions

1. Complete and deploy the continuous background scheduling workflow.
2. Conduct a controlled user evaluation with an appropriate sample size and ethical approval.
3. Measure reminder adherence, retention, cognitive workload, false-positive reminders, usability, and learner satisfaction.
4. Validate readiness estimates against reliable learner self-reports or other approved ground-truth measures.

5. Improve personalization using longer-term learner interaction histories.
6. Improve browser push reliability and deployment monitoring.
7. Evaluate scalability under realistic multi-user workloads.
8. Perform privacy, security, and accessibility audits before production deployment.

9.5 Final Statement

The developed component demonstrates a practical approach for integrating learner readiness and study progress into an adaptive reminder workflow. Its current contribution is a functionally implemented and tested prototype. With empirical evaluation and further engineering, the component can be developed into a more comprehensive learning-support service for university students and other digital-learning environments.

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GLOSSARY

Term	Definition
Adaptive reminder	A reminder whose timing or content is selected according to learner context rather than a fixed schedule.
Readiness	An estimate of whether the learner is currently in a suitable state for a study demand.
Retention probability	An estimated probability that a study item remains in memory.
Review priority	A value representing the urgency of revisiting a study item.
SM-2	A spaced-repetition algorithm that adjusts review intervals according to review history.
Reminder action	The selected scheduling outcome: SEND_NOW, DELAY, or SKIP.
Snooze	A learner action that postpones a reminder.
Dismiss	A learner action that rejects or closes a reminder.
Activity recommendation	The study activity suggested by the system for a particular learner/context.
Design Science Research	A research approach centred on designing and evaluating an artefact.

APPENDICES

Appendix A – System Screenshots

Insert only screenshots captured from your own running implementation. Recommended evidence includes:

- Reminder queue with due-today reminders.
- Upcoming/all-reminders view.
- Reminder settings and quiet-hour configuration.
- Reminder toast or browser notification.
- Reminder feedback action.
- Reminder decision/reason explanation.
- Study activity recommendation.
- Relevant backend/API response, with credentials and personal information removed.

Appendix B – Risk Analysis

Risk	Category	Impact	Mitigation
Incorrect retention estimate	Technical	Medium	Validate review behaviour and monitor review outcomes.
Readiness information unavailable	Integration	Medium	Use a defined fallback readiness mechanism.
Browser notification permission denied	Technical	Medium	Provide in-page reminder fallback.
Reminder overload	User experience	Medium	Support frequency controls and quiet hours.
Unauthorized reminder access	Security	High	Use authenticated learner-specific API access.

Sensitive learner-state information exposed	Privacy	High	Minimize stored data and protect access.
Incomplete integration with sibling modules	Integration	Medium	Use stable API contracts and test with mock inputs.
Insufficient empirical evidence	Research	High	Conduct controlled evaluation before making effectiveness claims.

Appendix C – Academic Integrity and AI Use Disclosure

Generative AI and other software tools were used for limited academic-support activities including document organization, language refinement, and formatting assistance. The author remains responsible for verifying the accuracy of the technical content, implementation claims, citations, testing evidence, screenshots, analyses, and conclusions. No empirical result should be treated as valid unless it is supported by the author's actual project evidence.